

GRADE 10 CBE NOTES

BIOLOGY

STEM PATHWAY

***THIS PDF COMPRISES OF PART OF THE NOTES OF THE NAMED
LEARNING AREA***

FOR COMPLETE GRADE 10 NOTES:

CONTACT

**MADAM PAULINE FOR COMPLETE
NOTES , SCHEMES LESSON PLANS
AND EXAMS 0726937493**

LIST OF LEARNING AREAS AT SENIOR SCHOOL

Compulsory Subjects	Science, Technology, Engineering & Mathematics (STEM)	Social Sciences	Arts & Sports Science
1. English 2. Kiswahili/KSL 3. Community Service Learning 4. Physical Education <i>NB: ICT skills will be offered to all students to facilitate learning and enjoyment</i>	5. Mathematics/Advanced Mathematics 6. Biology 7. Chemistry 8. Physics 9. General Science 10. Agriculture 11. Computer Studies 12. Home Science 13. Drawing and Design 14. Aviation Technology 15. Building and Construction 16. Electrical Technology 17. Metal Technology 18. Power Mechanics 19. Wood Technology 20. Media Technology* 21. Marine and Fisheries Technology*	22. Advanced English 23. Literature in English 24. Indigenous Language 25. Kiswahili Kipevu/Kenya Sign Language 26. Fasihi ya Kiswahili 27. Sign Language 28. Arabic 29. French 30. German 31. Mandarin Chinese 32. History and Citizenship 33. Geography 34. Christian Religious Education/ Islamic Religious Education/Hindu Religious Education 35. Business Studies	36. Sports and Recreation 37. Physical Education (C) 38. Music and Dance 39. Theatre and Film 40. Fine Arts

For Complete Grade 10 Senior schools; Notes, Textbooks, Teacher Guides, Revision Books, Schemes of work, Lesson Plans, Curriculum Designs, Records of work, Examinations, Assessment Materials, Practical Books and all the teaching and Learning materials,

BIOLOGY GRADE 10 SENIOR SCHOOLS

COMPLETE NOTES {STEM}

COMPETENCE BASED EDUCATION

BIOLOGY GENERAL LEARNING OUTCOMES

By the end of the course the learner should be able to:

1. Develop relevant knowledge, skills and attitudes for further education and for training in biology related scientific fields;
2. Demonstrate an understanding of interrelationships among humans, other organisms and their environment and apply the knowledge to conserve nature;
3. Describe features of various groups of living organisms and identify unknown organisms using simple biological keys;
4. Apply the knowledge gained on human body systems and functions to improve the quality of life for self and the community while enhancing healthy living;
5. Design and carry out practical activities and projects that will enable them to understand biological concepts;
6. Demonstrate relevant technical skills and scientific knowledge necessary for socio-economic development;
7. Demonstrate resourcefulness in designing projects necessary for community service learning;
8. Communicate biological information in a precise, clear and logical manner;
9. Apply knowledge on plant and animal structure and functions to industrialization, innovation and sustainability of life;
10. Use knowledge gained to make informed decisions about scientifically-based personal and societal issues to solve emerging issues in health and environment.

SUMMARY OF STRANDS AND SUB STRANDS

1.0 CELL BIOLOGY AND BIODIVERSITY

1.1 Introduction to Biology

1.2 Scientific investigations in Biology

1.3 Cell structure and Specialization

1.4 Chemicals of life

2.0 ANATOMY AND PHYSIOLOGY OF PLANTS

2.1 Nutrition

2.2 Transport

2.3 Gaseous exchange and Respiration

3.0 ANATOMY AND PHYSIOLOGY OF ANIMALS

3.1 Nutrition

3.2 Transport

3.3 Gaseous exchange and Respiration

STRAND 1: CELL BIOLOGY AND BIODIVERSITY

Sub-strand 1.1: Introduction to Biology

Definition of Biology - What is Biology?

Biology is a branch of science that deals with the study of living things.

- ✓ There are diverse forms of life on earth ranging from the invisible microscopic living things to the gigantic life forms. It aims at explaining the living world in terms of scientific principles.
- ✓ It is important to note, however, that living things interact with the non living things in the environment as well.
- ✓ Biology, therefore also entails the study of non living things as well. The role of human beings in shaping the environment is also investigated in biology.
- ✓ In summary, biology deals with the study of origins, types, nature, growth, development, interactions and maintenance of all life forms on earth.

Importance of biology

The study of biology is very important. The knowledge acquired from this study can benefit an individual in myriad ways. The study of biology is important in that:

- The knowledge acquired from the study of biology can be very helpful in solving environmental problems such as food shortage, poor health services, pollution and environmental degradation.
 - The study of biology can grant one an entry into various careers such as medicine, veterinary medicine, animal husbandry, horticulture and dentistry.
 - The study of biology leads to development of scientific skills which are very useful in life. These include skills of observing, identifying, recording, classifying, measuring, analyzing and evaluating. These skills can enable one learn how to make right choices and lead an improved life.
 - Through the study of biology man learns the causes of human, plant and human diseases and how best these diseases can be prevented and cured.
 - Biological knowledge acquired in the study of biology is very useful in enhancing international cooperation. Some biology related international conventions include:
 - Joint development of HIV/AIDS vaccine by Kenyan and British scientists.
 - The coordinated fight against Severe Acute Respiratory Syndrome involving scientist all over the world.
 - The fight to save the ozone layer from depletion through various international agreements such as the Kyoto protocol.
 - Management of resources through international treaties such as the CITES (Convention against International Trade on Endangered Species).
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Application of Biology in Everyday Life

Biology plays a role in:

<u>Area</u>	<u>Example of Application</u>
Medicine & Health	Disease diagnosis, drug development
Agriculture	Improving crop yield, livestock breeding
Environment	Conservation, waste management
Food Industry	Fermentation, food preservation
Forensics	DNA fingerprinting, crime solving Research & Tech Biotechnology, cloning, genetic engineering

Learner Activity:

- Learners work in groups to list how biology impacts their daily lives.
- Each group presents findings.

Fields of Study in Biology (15–20 min)

Branches of Biology

Biology is such a broad field of knowledge. It is divided into two broad branches.

1. Zoology - This is a branch of biology that deals with the study of animal life.
2. Botany- This is a branch of biology that deals with the study of plant life.

Within the two branches, there exist even smaller branches because the branches (botany and zoology) are very wide and complex.

Key Fields:

<u>Field of Biology</u>	<u>Description</u>
Botany	Study of plants
Zoology	Study of animals
Taxonomy	Classification of living things
Anatomy	Structure of living organisms
Physiology	Functions of body systems
Ecology	Relationships in ecosystems
Biochemistry	Chemical processes in organisms

Biotechnology	Application of biological systems
Genetics	Study of heredity and variation
Parasitology	Study of parasites
Microbiology	Study of microorganisms
Entomology	Study of insects

Learner Activity: □ Use textbooks or internet to research one field of biology.

- Share with the class: field name, what it involves, and at least one related career.

Careers Related to Fields of Biology (15 min)

<u>Field</u>	<u>Related Career(s)</u>
Botany	Botanist, Agricultural Officer
Zoology	Zoologist, Park Ranger
Genetics	Geneticist, Lab Technician
Biotechnology	Genetic Engineer, Biomedical Scientist
Ecology	Environmentalist, Ecologist
Anatomy	Surgeon, Physiotherapist
Microbiology	Microbiologist, Pathologist
Entomology	Entomologist, Pest Control Expert

Learner Activity:

- Role-play or create a skit showcasing a professional in a biology-based career.

Influences on Career Choices (10 min)

Positive Factors:

- Personal interest
- Abilities and talents
- Passion and motivation
- Career demand and job availability

Negative Factors (should NOT influence):

- Gender roles/stereotypes
- Cultural beliefs
- Disabilities
- Peer pressure
- Environmental background

Discussion Questions:

- “Why is it important to choose a career based on interest and ability?”
- “Can girls be surgeons or boys be florists? Why or why not?”

Assessment Questions

1. Define Biology.
2. Mention four careers that come from studying Biology.
3. Explain two ways Biology is useful in health and agriculture.
4. Motor vehicles move, use energy and produce carbon dioxide and water. Similar characteristics occur in living organisms yet motor vehicles are not classified as living. List the other characteristics of living things that do NOT occur in motor vehicles.

5. Give the name to the study of:

☐ The cell

☐ Micro—organisms

☐ The study of differences between parents and their offspring

☐ The study of relationships between organisms and their environment

C. Research/Project Work

- ☐ In pairs, research and create a poster showing one field of biology and three careers related to it. ☐ Present the poster to the class.

Sub-strand 1.12 scientific investigation in Biology

Collection of Specimen

We have defined biology as the study of living things. For effective study, a biologist may have to collect some living things or some parts of living things for observation and analysis. The living things or parts of living things that are used for biological study are called specimens. Biological studies always take place in laboratories. A laboratory is a building or a room that is designed and equipped for scientific studies. Collections of living things especially animals may not be very easy. Some of the animals are not easy to catch while some are quite dangerous. Knowledge on proper specimen collection and handling of is very important. We will discuss some of the apparatus used in specimen collection.

1. Apparatus for Specimen Collection and Preservation

<i>Apparatus/Material</i>	<i>Use</i>
<i>Pooter/Aspirator</i>	<i>Sucking small insects safely</i>
<i>Pitfall trap</i>	<i>Trapping crawling insects or animals</i>
<i>Soapy water</i>	<i>Used in traps for drowning insects</i>
<i>Forceps</i>	<i>Picking small specimens without damage</i>
<i>Sweep net / Aerial net</i>	<i>Catching flying insects</i>
<i>Light trap</i>	<i>Attracting and trapping nocturnal insects</i>
<i>Tullgren funnel</i>	<i>Extracting small animals from soil/litter</i>
<i>Envelopes</i>	<i>Holding butterflies or moths</i>
<i>Labels & Permanent Ink</i>	<i>Marking collected specimens</i>

<i>Hand lens</i>	<i>Magnifying small features</i>
<i>Tracing paper</i>	<i>Drawing and recording features</i>
<i>Gloves and Digger</i>	<i>Safe and effective soil specimen collection</i>
<i>Apparatus/Material</i>	<i>Use</i>
<i>Knife or Secateurs</i>	<i>Cutting plant parts</i>
<i>Collecting bags</i>	<i>Transporting specimens</i>

a. Sweep net

This is used for catching flying insects.



b. Fish net

This is used for trapping small fish and other small water animals.



c. Pooter

d.



This is used for sucking small animals from rock surfaces or barks of trees.

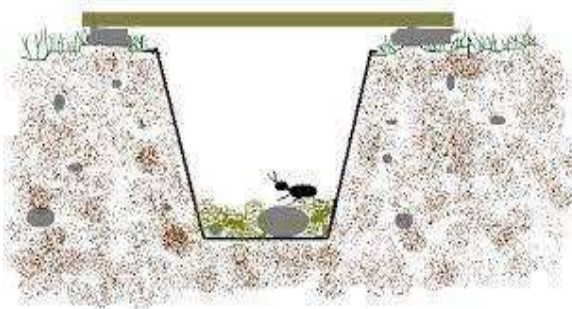
Bait trap

This is used for attracting and trapping small animals including rats.



e. Pit fall trap

This is used for catching crawling animals.



f. Pair of forceps

This is an apparatus used for picking up small crawling animals e.g. stinging insects.



g. Specimen bottles

These are bottles used for keeping collected specimen. They are of different sizes depending on the size of the specimen being studied.



h. Magnifying lens

This is used to enlarge small objects. A hand lens is a common magnifying lens used in the laboratory. The magnifying power of the hand lenses is always indicated on the lens e.g. X10, X5, X8. The magnifying power of a lens shows how many times the image will be enlarged compared to the object.

LIST OF GRADE 10 SENIOR SCHOOLS MATERIALS AVAILABLE:

1	TEACHING/LEARNING NOTES
2	KICD APPROVED TEXTBOOKS
3	KICD APPROVED TEACHER GUIDES
4	REVISION BOOKS
5	SCHEMES OF WORK
6	LESSON PLANS
7	KICD APPROVED CURRICULUM DESIGNS
8	RECORD OF WORK
9	EXAMINATIONS & ASSESSMENT MATERIALS
10	APPROVED PRACTICAL BOOKS & PRACTICAL MATERIALS
12.	APPROVED LITERATURE & FASIHI BOOKS & ALL RELATED MATERIALS